

Wi-Fi Transmission Over In-building FTTR Networks Using Sigma-Delta Modulation

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Abstract *Sigma-delta, unlike analog radio over fiber, enables robust Wi-Fi FTTR transmission with standard SFP+ modules not optimized for linear drive. Furthermore, a 27.5% reduction in required optical bandwidth is achieved for a 160 MHz 256-QAM signal by reducing the oversampling rate through amplitude clipping. ©2026 The Author(s)*

Introduction

The combination of fiber-to-the-room (FTTR) with Wi-Fi is an enabler for high-performance wireless indoor access in households and industries. The most popular FTTR solutions today implement an optical distribution network (ODN) either upon GPON, operating natively at symmetrical 2.5 Gbps or at symmetrical 10 Gbps with XGS-PON. A central Main FTTR Unit (MFU) interfaces with the external access network and acts towards so-called Sub FTTR Units (SFU) in each room as an optical multiplexer and controller. Both MFU and SFU integrate Wi-Fi access points by converting the optical signals to baseband and re-transmitting the data over the air as Wi-Fi frames for downlink and in-reverse for uplink.

FTTR now evolves from essentially a secondary PON layer, cascaded with classical access PON, to an ODN tailored more to the in-building context, with MFU-centralization of Wi-Fi functionality being a key trend [1-3]. Its most consequential variant is analog radio over fiber (ARoF), where Wi-Fi signals are distributed with the ODN, reducing the functionality at SFUs to analog baseband-passband conversions. This vastly simplifies the SFUs and requires only a minimal ODN bandwidth. However, highly linear optics are a prerequisite and higher point-to-multipoint split ratios become difficult due to reflections.

An alternative variant is sigma-delta radio over fiber ($\Sigma\Delta$ -RoF). The sigma-delta principle relies on oversampling and quantization with a low amplitude resolution. A negative feedback loop spectrally shapes the quantization noise and

shifts it towards higher frequencies outside of the signal bandwidth. Subsequent lowpass filtering removes this high frequency noise, enabling highly accurate amplitude reconstruction in the baseband [4]. Sigma-delta-based optical transmission therefore combines key advantages of both ARoF and digital IQ transport. It removes constraints on linearity and resolution by trading amplitude resolution, down to 1 bit, for a significantly increased sampling rate. From an FTTR system perspective, $\Sigma\Delta$ -RoF enables an extremely simple SFU design, since all processing is shifted to the central unit. It also allows the use of lowest-cost optical and electrical components and an all-digital transmitter [5,6]. The main drawback of $\Sigma\Delta$ -RoF is the increased optical bandwidth requirement due to oversampling-induced spectral expansion, which may limit scalability in bandwidth-constrained FTTR deployments.

In this work and for the first time to our knowledge, we experimentally demonstrate the transmission of Wi-Fi signals, complying with the latest IEEE 802.11 be standard [7] over an FTTR link, comparing ARoF and $\Sigma\Delta$ -RoF. We investigate both, a low-cost SFP+ and a premium module using a linear drive VCSEL and evaluate the impact of waveform clipping as a means to enhance system performance.

Experimental Setup

The experimental setup for the transmission of Wi-Fi signals for an FTTR link can be found in Fig. 1. As the first step of the digital signal processing (DSP), at the transmitter the Wi-Fi

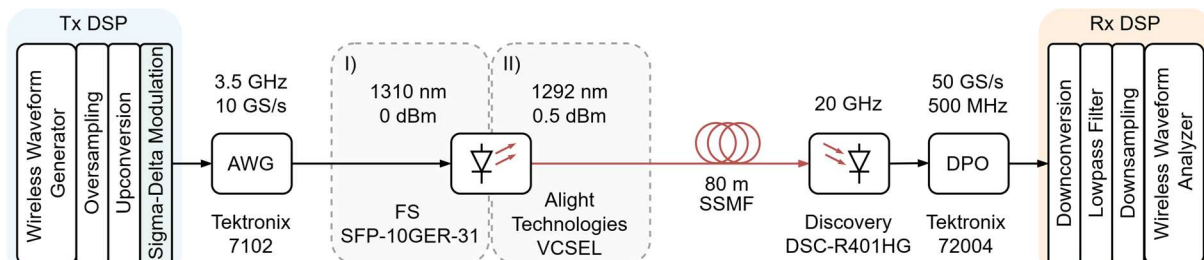


Fig. 1: Experimental setup of the Wi-Fi radio over fiber transmission with either an SFP+ or a VCSEL as the optical transmitter.

waveform is generated in MATLAB using the WLAN Toolbox [8]. The channel bandwidth is chosen as 160 MHz or 320 MHz and the modulation and coding scheme (MCS) as 6 or 8, corresponding to a 64- or 256-quadrature amplitude modulation (QAM) with a coding rate of 3/4, resulting in a waveform that is compliant with the IEEE 802.11be standard ("Wi-Fi 7"). The generated waveform is then oversampled with the oversampling ratio OSR. Following, the signal is upconverted by mixing with an intermediate frequency f_{IF} of 85 MHz or 175 MHz for the 160 MHz or 320 MHz signal, respectively. The signal is subsequently processed by a second-order sigma-delta modulator. Within the modulator, the input signal is quantized using a 1-bit quantizer, resulting in a PAM-2 representation. The quantization noise is shaped around the center frequency f_{IF} of the target band [4]. By increasing the quantizer resolution, reduced quantization error levels and higher-order PAM constellations, can be achieved. For an ARoF transmission the sigma-delta modulator is omitted and the upconverted signal is directly sent to the arbitrary waveform generator (AWG) (Tektronix 7102). After the AWG the signal is converted to the optical domain utilizing either an enhanced small form-factor pluggable (SFP+) transceiver or a vertical-cavity surface-emitting laser (VCSEL) from Alight Technologies and fed into 80 m of standard single-mode fiber (SSMF). The signal is detected by a photodiode (PD) with a transimpedance amplifier (Discovery DSC-R401HG) and subsequently recorded using a digital phosphor oscilloscope (DPO) (Tektronix 72004). A software-based bandwidth limitation to 500 MHz is applied in order to suppress out-of-band noise. After digitization, the signal is downconverted by mixing with the intermediate frequency f_{IF} . After this, a rectangular lowpass filter with the cutoff at the signal bandwidth of either 160 MHz or 320 MHz is applied before the signal is downsampled. The WLAN Toolbox is used for the evaluation of the waveform. This includes all required Wi-Fi processing and MMSE equalization, where the equalizer operates in the frequency domain with one complex coefficient per active

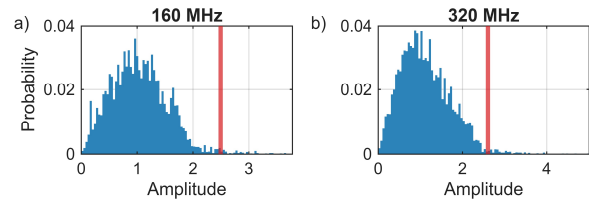


Fig. 2: Probability distribution of Wi-Fi signal amplitude levels and optimal amplitude clipping thresholds.

subcarrier, resulting in 1960 and 3920 MMSE weights for the 160 MHz and 320 MHz signal, respectively. Finally, the resulting error vector magnitude (EVM) is evaluated. For the selected MCS 6 or 8, the standardized EVM requirement is 4.5% or 3.2% [7].

Clipping of Waveform

The waveforms in Fig. 2 illustrate the well-known peak-to-average power ratio (PAPR) challenge of OFDM systems. Larger amplitude values occur with significantly lower probability. Here, the PAPR is 11.6 dB for the 160 MHz signal. A higher signal bandwidth of 320 MHz leads to a higher maximum amplitude value and therefore a higher PAPR of 14.1 dB. This behavior directly translates to the PAM representation of the signal after sigma-delta modulation, with increasing severity for higher orders, as the occurrence of the outermost levels is less probable and the advantage of gained resolution is not fully exploited.

To mitigate this effect, the straight-forward PAPR reduction technique hard clipping is introduced, where all amplitude values exceeding the optimal threshold are clipped, which reduces the PAPR to 9.6 dB and 10.5 dB for the 160 MHz and 320 MHz signal. More advanced PAPR reduction techniques could reduce out of band signal leakage, but are out of scope for this paper [9].

Results for SFP+ Measurements

First, transmission using a standard SFP+ transceiver was investigated. For this case, the applied clipping factor was optimized based on the EVM for the respective signals as shown in Fig. 3a). For the 160 MHz signal, the optimal clipping factor is 0.66 (9.6 dB PAPR). The minimum EVM for the 320 MHz signal is observed at a clipping factor of 0.52 (10.5 dB PAPR). The shift to the left of the optimum is attributed to the higher

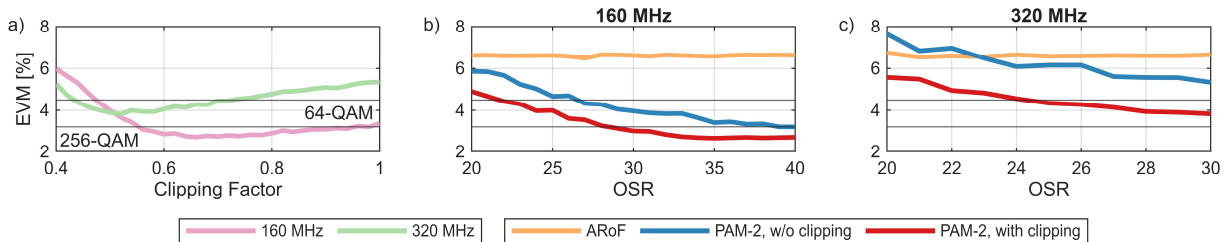


Fig. 3: a) Optimization of clipping factor for a 160 MHz and 320 MHz Wi-Fi signal. b) Measured EVM versus OSR for ARoF and PAM-2 with and without clipping applied for a 160 MHz Wi-Fi signal. c) Measured EVM versus OSR for ARoF and PAM-2 with and without clipping applied for a 320 MHz Wi-Fi signal.

peak amplitude as previously discussed and shown in Fig. 2 resulting in a smaller optimal clipping factor and therefore a stronger clipping.

The optimized clipping factors are then applied to the respective signals and the performance is compared to both ARoF transmission and the unclipped case. The EVM versus OSR for the 160 MHz signal is shown in Fig. 3b). ARoF transmission is not feasible as the threshold is not met, which can be attributed to the strong nonlinearities introduced by the optical transmitter. This clearly highlights the advantage of $\Sigma\Delta$ -RoF transmission under pronounced nonlinear distortion. The unclipped signal reaches below the 4.5% threshold for 64-QAM at an OSR of 27, and below the 256-QAM threshold of 3.2% at an OSR of 40. This corresponds to a required optical transmission rate of 6.4 GBd. In comparison, the optimized and clipped signal meets the threshold at an OSR of 29, corresponding to a required optical rate of 4.64 GBd, resulting in a rate reduction of 27.5% when clipping is applied.

For the 320 MHz signal seen in Fig. 3c) the impact of waveform clipping is even more significant. Neither the ARoF signal nor the unclipped PAM-2 sigma-delta modulated signal achieve an EVM below the 4.5% threshold. However, with optimized clipping the PAM-2 sigma-delta signal meets the threshold at an OSR of 25, corresponding to a required optical transmission rate of 8 GBd.

Results for VCSEL Measurements

Next, the performance of a VCSEL-based transmission of a 320 MHz signal was investigated. As in the previous section, the clipping factor was first optimized to identify the optimal operating point, as seen in Fig. 4. Due to the more linear behavior of the VCSEL, the use of higher-order PAM sigma-delta modulation becomes feasible. Furthermore, the performance of ARoF is no longer significantly degraded and the advantages of sigma-delta modulation are no longer observable under these conditions. For ARoF transmission, the optimal clipping factor is found to be 0.62 (11.9 dB PAPR). For PAM-8 and PAM-6 sigma-delta modulated signals, the optimum shifts to 0.56 (11.1 dB PAPR) and for PAM-4 and PAM-2 to 0.52 (10.5 dB PAPR). At these

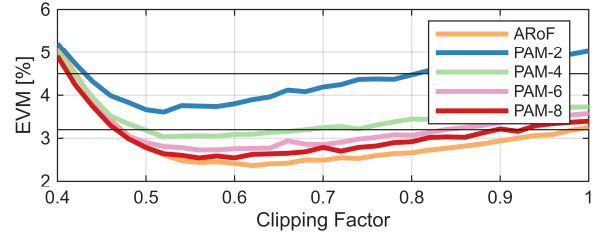


Fig. 4: Optimization of the clipping factor for a 320 MHz Wi-Fi signal for analog and sigma-delta radio over fiber.

operating points, all minima lie below the 4.5% EVM threshold, and all cases except the PAM-2 sigma-delta modulated signal also satisfy the more stringent threshold of 3.2% required for 256-QAM. Fig. 5 shows the improvement in EVM for the aforementioned 320 MHz signals when optimized clipping is applied, compared to the unclipped case. For the case of ARoF transmission shown in Fig. 5a), the curve remains below the 3.2% threshold for all OSRs. The PAM-2 signal reaches the 4.5% threshold at an OSR of 25, as illustrated in Fig. 5b). The PAM-4 signal in Fig. 5c) exhibits a rate reduction of 12.5% by shifting from an OSR of 24 to 21 at the 4.5% EVM threshold, while meeting the 3.2% threshold at an OSR of 29. The PAM-6 signal in Fig. 5d) achieves a reduction of 22.7% at the 4.5% threshold and satisfies the 3.2% threshold at an OSR of 25. For the PAM-8 signal, the required transmission rate is reduced by 25% at the 4.5% threshold when clipping is applied and the 3.2% EVM threshold now being met at an OSR of 21, corresponding to an optical transmission rate of 6.72 GBd, as shown in Fig. 5e). Overall, the performance advantage provided by waveform clipping becomes more pronounced with increasing PAM order, as the impact of the modified amplitude level distribution becomes more significant.

Conclusions

We demonstrate $\Sigma\Delta$ -RoF as the preferred transmission scheme for Wi-Fi signals in FTTR applications where strong component nonlinearities are present as ARoF becomes infeasible. Waveform clipping significantly enhances performance further, enabling an OSR and subsequent optical bandwidth reduction of 27.5% for 160 MHz Wi-Fi signals at 256-QAM. The impact becomes more pronounced for higher channel bandwidths and higher-order PAM sigma-delta modulation.

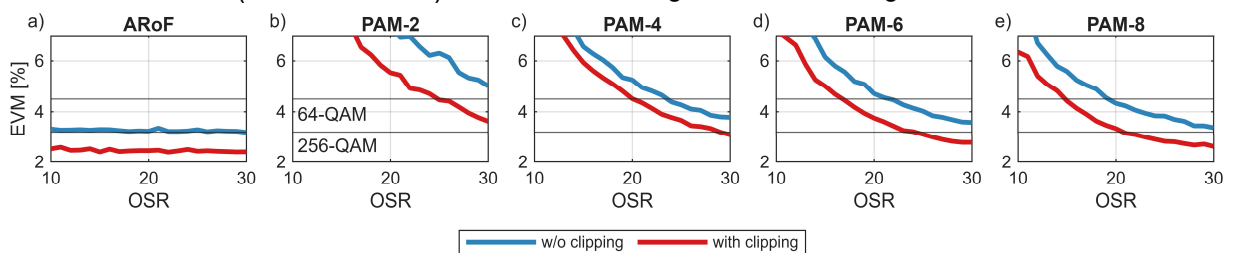


Fig. 5: Measured EVM versus OSR for analog and sigma-delta radio over fiber with different PAM orders with and without clipping applied for a 320 MHz Wi-Fi signal.

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